Learn Geospatial Correlations for Remote Sensing Image Change Detection

Jia Liu, *Member, IEEE*, Jingxuan Ju, Wenhua Zhang, *Member, IEEE*, Fang Liu, *Senior Member, IEEE*, Jingxiang Yang, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*—Change detection (CD) plays a critical role in Earth's surface surveillance via remote sensing (RS) images, which usually have strong geospatial correlations in surface entities, including specific global layout of different scenes and contextual dependencies of local regions or objects. To capture such geospatial correlations and facilitate recognition of changed regions, based on an attention mechanism, we propose a geospatial correlation feature enhancement network (GCFENet) for RS image CD. In the network, to capture global scene context and change details, a dual feature interaction (DFI) module is first designed. Then, to model and fuse different correlations for each scale, a three-branch multicorrelation attention (MCA) block is proposed, which models local distributions, global scene dependencies, and pixelwise semantic prior, respectively, and associates them via a self-attention module to capture the semantic distributions within the scene and between different regions. In the MCA network, a Gaussian filter-based frequency separation (GFS) module is designed to generate the global scene dependencies and local distributions, which are used as the value and the query in self-attention, respectively. Then, a local semantic modeling (LSM) module is proposed to assign pixelwise semantic prior, which is used as the key to generate correlations with the query and the value. This GCFENet enables the attention mechanism to compute similarity not only based on pixelwise feature matching but also by perceiving the geospatial distribution correlations of ground objects and bi-temporal spatial correspondences. Experiments on three public datasets demonstrate the state-of-the-art performance of the GCFENet. The code will be open-sourced in [https://github.com/4043-lab/Change-detection] once accepted.**

***Index Terms*—Attention mechanism, change detection (CD), remote sensing (RS), semantic prior.**

I. Introduction

CHANGE detection (CD) is a crucial research direction in remote sensing (RS) images, aiming to identify the ground-object changes in the same area over different time periods, which has been widely applied in diverse fields, such as land management [1], [2], disaster assessment [3], [4], environmental monitoring [5], [6], and resource management [7].

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The authors are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: whzhang@njust.edu.cn).

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As a core realm in RS, traditional CD methods have evolved from pixelwise to feature-driven approaches over the past decades. Constrained by early sensor limitations, pixelwise algebraic operations dominated the early era: directly generating difference maps through image differencing [6] or ratio operations [8], followed by segmentation for change identification. With medium- to high-resolution satellite development, transformation-based feature-level methods replaced pixelwise analysis: principal component analysis (PCA) [9] and tasseled cap transformation (KT) [10] converted spectral data into physically meaningful features for change discrimination. When technical demands advanced to change-type identification, classification-based methods emerged. Support vector machines (SVMs) [11] and random forests (RFs) [12] analyzed object category variations for pixelwise CD, but struggled with high-resolution images' massive pixel data and complex feature processing challenges. In recent years, deep learning (DL), as a revolutionary tool, has deeply influenced the fields of computer vision and RS. Consequently, an increasing number of DL-based CD algorithms have been gradually developed, with their powerful feature learning and complex model construction capabilities significantly enhancing the effectiveness of CD. Meanwhile, DL models can automatically learn abstract feature representations from datasets, eliminating the need for manually designing feature extraction rules and greatly improving the ability to capture change information in complex scenes. Initially, deep convolutional neural networks (CNNs) played a crucial role. For example, Daudt et al. [13] introduced the fully convolutional network (FCN) [14] into the field of CD. Based on U-Net [15], they designed FC-Siam-Conc by directly concatenating bi-temporal images for detection; FC-EF by first fusing features of bi-temporal images and then processing them; and FC-Siam-Diff by calculating feature differences. Among these, the approach of taking bi-temporal images as an integrated input set a precedent, serving as the foundation for most subsequent networks [16], [17], [18]. Later, Peng et al. [19] developed U-Net++ based on a multiscale feature fusion enhancement strategy, while Fang et al. [20] proposed SNUNet using spatiotemporal enhanced fusion. Despite the good application effects demonstrated by these methods, due to the inherent limitation of CNNs, the limited receptive field, these networks struggle to effectively model global information, leading to insufficient recognition accuracy for objects in complex scenes. Therefore, researchers have attempted to introduce the attention mechanism of vision transformers into the field of RS image CD [21], [22], [23] to address the issue of insufficient receptive field in CNNs. For example, Chen et al. [24] designed the bi-temporal image transformer (BIT), which uses transformers

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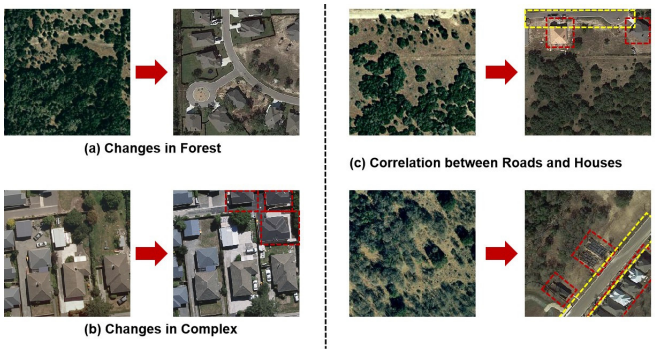


Fig. 1. Illustration of geospatial correlations in RS CD. (a) and (b) Cross time change patterns. For example, the buildings are usually changed from forest or farmland, as in (a), while seldom from a building complex, as in (b). (c) Dependence relationships between changed objects, such as the roads and houses.

to achieve spatiotemporal alignment, while Bandara and Patel [25] specifically designed a siamese transformer architecture for RS CD, termed ChangeFormer. The introduction of the attention mechanism enables effective modeling between any region in the image and the entire image [26], [27].

Although the above-mentioned DL-based methods have achieved remarkable results, they seldom specifically consider the geospatial correlations of the global scene or between regions. The changes of features in RS images do not occur independently, but follow geospatial patterns.

To effectively capture such patterns, both local and global perspectives are essential. Local distributions describe adjacency and co-occurrence relationships between neighboring surface objects (e.g., roads and buildings), which help reveal short-range dependencies. Meanwhile, global scene dependencies encode the overall spatial layout and functional composition of the region, reflecting long-range contextual dependencies. Modeling these two complementary aspects jointly allows the network to approximate the intrinsic geospatial dependencies of surface entities and their temporal evolution. As shown in Fig. 1(a) and (b), different scenes exhibit distinct change patterns: large areas of forests are cleared for construction, while complexes rarely undergo changes. On the other hand, the local distribution of features also demonstrates interdependence, as shown in Fig. 1(c), roads and houses often appear in pairs. If we can properly grasp such geospatial correlations of features and the logic of human activities on the basis of previous research, it will significantly aid the CD task in RS images. To achieve this goal, the network should extract global scenes, as well as pixelwise regional semantics, and associate them interactively.

Inspired by such geospatial patterns and the above analysis, this article proposes the geospatial correlation feature enhancement network (GCFENet). It follows the Siamese framework to first extract bi-temporal multiscale features. Following this motivation, the dual feature interaction (DFI) module first enhances fine-grained change details and global contextual information. Then, a three-branch multicorrelation attention (MCA) block is designed, which extracts global scene dependencies, local distributions, and semantic priors from the enhanced feature and captures the geospatial relationships among them via the attention mechanism. In MCA, the local semantic modeling (LSM) module is proposed to generate a semantic prior (Key), which serves as intermediate perceptual elements in self-attention to calibrate spatial coordinates and balance the deepened local distributions (Query) and global scene dependencies (Value). These local distributions and the global scene dependencies are obtained via the proposed Gaussian filter-based frequency separation (GFS) module. Finally, the prediction head completes the integration of multiscale feature maps to obtain the final CD prediction map. The main contributions of our work are summarized as follows.

1) We propose the GCFENet to capture the geospatial correlations of RS images for CD. It achieves this goal via feature interaction modules and specially designed self-attention blocks.
2) A DFI module is proposed to first enhance the change details and global scene context in order to assist the learning of geospatial correlations.
3) We propose the MCA block, which extracts pixelwise semantic prior, local distributions, and global scene dependencies, respectively, and associates them via a self-attention mechanism.
4) Experiments show that GCFENet enables the model to learn the geospatial correlations between times and among ground objects in RS images. Furthermore, qualitative and quantitative studies on three datasets show the state-of-the-art performance of GCFENet.

The remainder of this article is structured as follows. Section II provides a concise review of related work, highlighting key advancements and challenges in the field. In Section III, we delve into the architectural components and design specifics of the proposed network, detailing its layered structure, operational mechanisms, and innovative design choices. Section IV is dedicated to experimental comparisons, where we systematically evaluate the performance of our approach against state-of-the-art methods using benchmark datasets and quantitative metrics. By conducting ablation experiments, we systematically investigate in Section V the specific performance of each module of GCFENet. Finally, Section VI synthesizes the findings and concludes this article, outlining future research directions inspired by this work.

II. Related Works

The proposed method focuses on multifeature learning and its association via attention mechanisms. In this section, we provide a concise review of feature-based CD methods and attention mechanisms in CD.

A. Feature-Based CD Methods

Zhan et al. [28] proposed a Siamese CNN for optical aerial image CD, providing an effective solution for automated CD in the field of RS. Since then, the Siamese networks have been one of the mainstream frameworks for CD. Daudt et al. [13] proposed an FCN based on U-Net for CD. Subsequently, a large number of excellent networks emerged, such as the well-known U-Net++ [19]. To detect changes, spatial–temporal features are critical; thus, Chen et al. [29] proposed a continuous cross-resolution architecture, integrating the spatiotemporal features of multiresolution RS images to achieve high-precision detection. To generate accurate changed regions, local or structural details are important

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information. For example, Guo et al. [30] extracted pixelwise differences using metric learning, and Noh et al. [31] proposed a single-temporal image reconstruction loss to capture structural changes and further combined with the segment anything model to refine rough change regions. Recently, multiscale features have been popular to simultaneously represent both detail and semantic information. For example, the ChangeFormer model [25] extracts multilevel features, samples them, normalizes them to a unified scale, and fuses them. Hou et al. [32] fused multiscale feature differences of bi-temporal phases through a symmetric network. SEIFNet [33] enhances the feature representation capability for change regions by fusing multilevel feature maps.

While existing methods seldom consider the geospatial correlations as introduced above, in this article, we extract multiscale features based on the Siamese framework and try to capture such correlations via specially designed attention blocks.

B. Attention Mechanisms in CD

With the development of vision transformers, attention mechanisms have been widely applied in CD to model the entire image. For example, IFNet [16] combines attention mechanisms and spatial attention for feature fusion. SwinSUNet [21], based on Swin Transformer, adopts a Siamese U-shaped structure and a hierarchical sliding-window self-attention mechanism, especially dealing with the change feature extraction and localization of bi-temporal RS images. Feng et al. [23] designed the JoinAtt module to achieve feature interaction, making the attention focus on changes. Based on the hybrid attention mechanism of the transformer, Li et al. [34] completed the extraction of spatiotemporal contextual information of bi-temporal features and designed DARNet. Song et al. [35] used axial cross-attention to achieve feature enhancement. BIT [24] uses a dual-protagonist mechanism to dynamically capture the global context of images and cross-temporal feature interactions. In BiFA, Zhang et al. [36] introduced a bi-temporal cross-attention mechanism. The above models make full use of attention mechanisms to handle the problems of feature alignment and context modeling in CD. However, they often ignore the specific patterns of the functional coupling and survival dependence relationship of the ground object distributions in RS images due to the routines of human activities, as shown in Fig. 1.

In this article, we focus on the geospatial correlations and propose the MCA block to associate local and global correlations for fused features based on the self-attention module.

III. Methodology

In this section, we first introduce the comprehensive architecture of GCFENet. We then provide a detailed explanation of the design rationale behind DFI. Regarding the MCA module, we first present a high-level overview of its overall structure, followed by detailed descriptions of the LSM and GFS submodules in its branches. Finally, we discuss the loss function that we have devised for GCFENet.

A. Architecture Overview

The architecture of the proposed GCFENet is illustrated in Fig. 2. For input bi-temporal RS images T_1 and T_2, a shared-weight Siamese backbone (ResNet18) first extracts features, generating four sets of multiscale feature maps at different stages. These feature maps are processed in groups through the DFI, which integrates local–global contextual information to derive initial bi-temporal difference features at each scale. Subsequently, the difference maps are fed into the MCA. This component incorporates scene-level global representations and local distribution patterns into the features. In the three branches of MCA, in addition to achieving approximate "local" and "global" representations by varying the timing of partition operations, we have designed two supplementary modules: LSM and GFS. The LSM module assigns a semantic prior per pixel, allowing a preliminary segmentation of "changed or unchanged" regions. The GFS module is incorporated into two of MCA's branches; it leverages principles from digital image processing and utilizes a symmetric, lightweight filtering algorithm to further enhance both local and global features. Following MCA is the attention module: in the attention mechanism's computation, the semantic prior (K) serves as intermediate perceptual elements, which link the local distributions (Q) with the global scene dependencies (V). Finally, the reassembled feature maps are upsampled to the same spatial resolution and fused with learnable stagewise scalar weights, after which the prediction head produces the final map. Specifically, each stage i is associated with a trainable scalar w_i that is optimized jointly with the network; a softmax normalization across stages enforces nonnegative weights with $\sum_{i=1}^{4} w_i = 1$. The fused representation is computed as

$$F_{\text{fuse}} = \sum_{i=1}^{4} w_i F_i \tag{1}$$

which allows the model to adaptively balance the contributions of different semantic levels rather than assuming equal importance.

B. Dual Feature Interaction Module

CNNs and attention mechanisms represent two dominant architectural paradigms currently in use, each with distinct advantages and limitations. Constrained by their receptive fields, CNNs excel at capturing local contextual information, whereas attention mechanisms demonstrate superior capabilities in modeling global dependencies. To fully leverage the complementary strengths of these two approaches, when designing dual-phase image difference feature maps, we adopt a dual-branch architecture, integrating CNN-based and attention-based modules, respectively, as illustrated in Fig. 3.

The DFI consists of additive and subtractive branches. The additive branch improves semantically consistent features by pixelwise summation, emphasizing unchanged regions in CD, typically large-scale areas that carry global scene context. This branch employs the convolutional block attention module (CBAM) [37] to effectively model global contextual information. Mathematically, this process can be expressed as

$$F_{add}^{i} = \text{CBAM}\left(F_1^i \oplus F_2^i\right) \tag{2}$$

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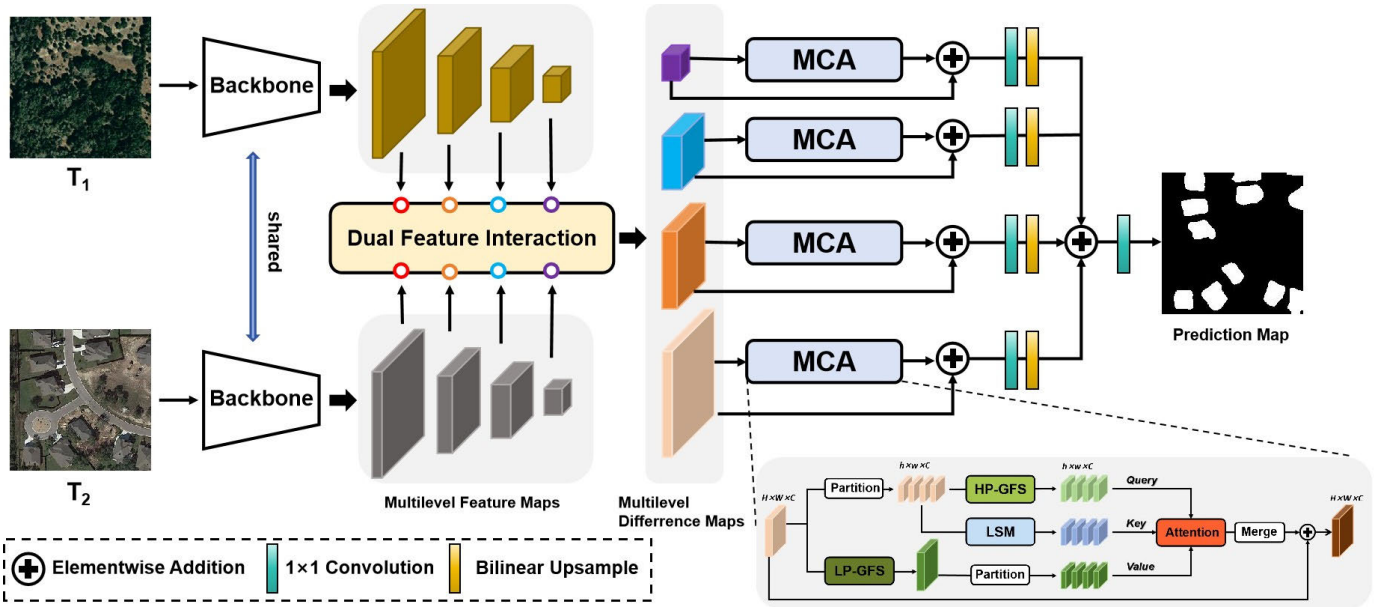


Fig. 2. Architecture of GCFENet.

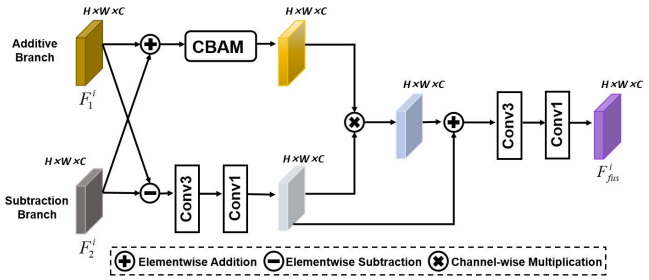


Fig. 3. Illustration of the DFI.

where F^i_{add} represents the ith feature map output by the additive branch, F^i_1 and F^i_2 denote the input features from time T_1 and T_2, respectively, and $\oplus$ represents the elementwise addition. The CBAM further refines the summed features through channel and spatial attention mechanisms

$$\text{CBAM}(x) = \text{Spatial}(\text{Channel}(x)) \tag{3}$$

where the channel attention Channel$(\cdot)$ captures interdependencies between feature maps, and the spatial attention Spatial$(\cdot)$ highlights region-specific contextual cues. This design ensures that the additive branch selectively amplifies global scene context critical for detecting large-scale unchanged areas.

The subtractive branch computes pixelwise absolute differences between bi-temporal feature maps to highlight regions of change. This operation effectively captures sparse and irregularly distributed changes, which are then processed by a CNN to model their spatial dependencies. Mathematically, this process is formulated as

$$F^i_{sub} = \text{Conv1}\left(\text{Conv3}\left(|F^i_1 - F^i_2|\right)\right) \tag{4}$$

where F^i_{sub} represents the ith feature map output by the subtractive branch; F^i_1 and F^i_2 denote the ith feature maps of two temporal phases, respectively; $|\cdot|$ indicates the elementwise absolute value operation for calculating the absolute difference between bi-temporal feature maps; Conv3 and Conv1 represent the convolutional operations with specific kernel configurations (3×3 and 1×1), respectively, which are used to capture sparse and irregularly distributed change details.

Subsequently, through pixelwise multiplication, the "attention weight map" F^i_{add} generated by the addition branch is used to weight the change details F^i_{sub} refined by the subtraction branch in both channel and spatial dimensions, emphasizing change details with global context. To enhance robustness, a residual connection operation is then applied. Finally, feature enhancement is performed via CNN operations

$$F^i_{out} = \text{Conv1}\left(\text{Conv3}\left(F^i_{add} \odot F^i_{sub} + F^i_{sub}\right)\right) \tag{5}$$

where $\odot$ denotes the multiplication by element.

From a theoretical perspective, the additive operation reinforces feature activations that remain consistent across the bi-temporal pair, thereby suppressing temporal noise and highlighting unchanged regions. These invariant responses serve as a stable semantic reference for subsequent attention modeling. Although both the DFI and MCA modules incorporate local and global contextual cues, DFI performs feature-level decomposition to separate invariant and variant components, while MCA, introduced in Section III-C, operates at a higher semantic level to model multiscale correlations, thus avoiding functional redundancy.

C. Three-Branch MCA Block

The MCA block reconstructs the input x to the attention mechanism through different structural transformations with distinct objectives across branches. This design enables the attention mechanism to incorporate the unique land cover correlations inherent in RS imagery during computation, thus enhancing the model's capability to detect change regions.

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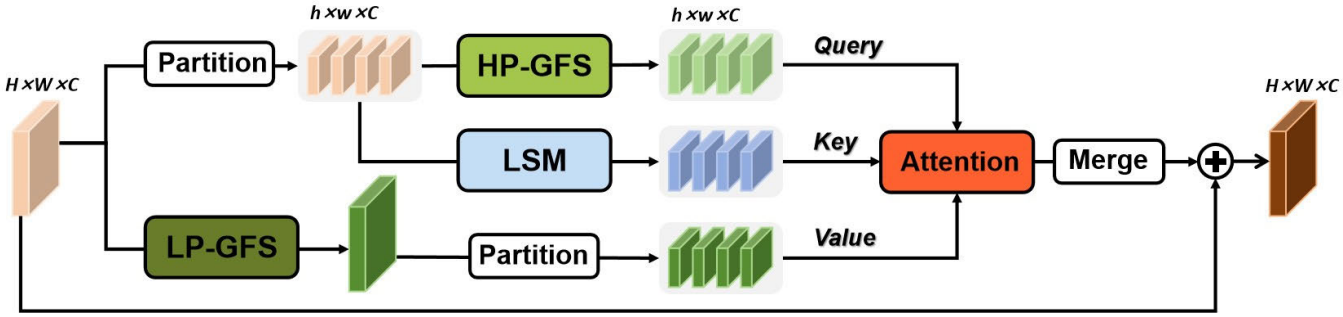


Fig. 4. Illustration of the MCA.

Specifically, changes in RS imagery are influenced by the logic of human activity and temporal evolution patterns.

We will first present an overview of the overall architecture of MCA, followed by the detailed design of the LSM and GFS modules within MCA.

1) Overall Architecture of MCA: As illustrated in Fig. 4, the MCA consists of three branches. In the Query branch, the feature maps are first subjected to partition operations, and each small feature map is processed by the high-pass GFS (HP-GFS) module to extract local distributions. In the Key branch, semantic priors are incorporated pixelwise through the LSM. In the Value branch, the low-pass GFS (LP-GFS) module is first employed to obtain the global scene dependencies, followed by partition operations to balance the input to the attention module. Finally, the blocks output by the attention module undergo reassembly and residual addition operations to yield the final result.

Adjacent objects in RS imagery often exhibit spatial–temporal dependencies—for example, high-rise buildings are typically constructed alongside roads, and container terminals are followed by freight warehouses. To capture such local relationships, we partition the feature map into uniformly sized patches and apply the HP-GFS module to perform Gaussian high-pass (HP) filtering, thereby enhancing fine structural details. Although no explicit coordinate encoding is used, the local spatial dependence is implicitly embedded in the Query branch through the patch partition and HP filtering, ensuring that the attention mechanism attends to adjacent object regions during subsequent computations. The filtered output is then used as the Query (Q) in the attention mechanism.

The Value (V) branch is designed to capture global scene dependencies, which are crucial given the distinct evolutionary patterns across different RS scenarios. For example, large areas of vacant land can evolve into building clusters, whereas such transformations are unlikely in oceanic regions. Therefore, inspired by the principle of symmetric design, in this branch, we introduce a Gaussian low-pass (LP) filtering design. The unpartitioned full-resolution feature map is first processed by the LP-GFS module to impose a form of global scene dependencies. To ensure compatibility with subsequent operations, the resulting feature map is then partitioned into patches.

In addition, the Key (K) branch integrates scene-specific semantic prior through the LSM after partition operations. Leveraging predefined semantic channels, this branch adaptively adjusts feature weights based on the predicted likelihood

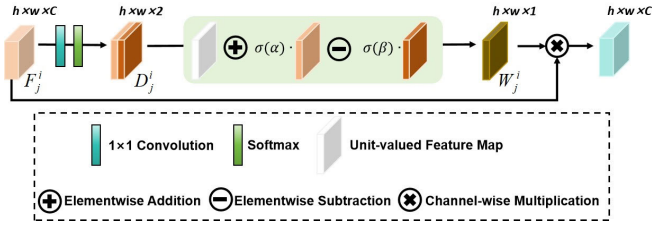


Fig. 5. Illustration of the LSM.

of change. Specifically, pixelwise features are reweighted to emphasize regions with great change probability while suppressing background noise, thereby enhancing model robustness against complex contextual variations.

Through these three branches, our framework injects distinct yet complementary enhancements to the input features x prior to the QKV projection. The subsequent attention mechanism inherently facilitates interactions between the scene-level global representations (captured by the V branch) and object-specific local spatial distributions (modeled by the Q branch) while leveraging a semantic prior (encoded in the K branch) as an intermediary perception layer. This design is particularly crucial for modeling land-cover dependencies in RS imagery: within a vast forest, the presence of a trail often indicates subsequent construction of a cabin; similarly, sparse human activities on vacant land can foreshadow large-scale residential development during urbanization.

2) LSM Module: RS imagery is inherently susceptible to complex backgrounds, weather-induced illumination variations, and seasonal discrepancies, which can introduce substantial noise when directly fusing local features with global context. To address this challenge, we propose an LSM that embeds binary semantic labels indicating "changed or unchanged" states into pixelwise features. By leveraging these spatial prior masks as an intermediate layer, our framework effectively balances the interaction between local distributions and global scene dependencies, mitigating the impact of background clutter on CD performance.

As illustrated in Fig. 5, the bi-temporal difference feature map F^i is first partitioned into nonoverlapping blocks F_j^i. Each block undergoes a preclassification operation via a 1×1 convolutional layer to project the multidimensional features onto a two-channel space. A softmax operation is applied per channel to generate the preclassification map D_j^i, where the two channels represent the probability distributions of

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unchanged and *changed* regions, respectively. The semantic prior weight matrix W_j^i is computed as

$$W_j^i = 1 + \sigma(\alpha) \cdot D_j^{i(changed)} - \sigma(\beta) \cdot D_j^{i(unchanged)} \tag{6}$$

where α and β are the learnable scalar parameters optimized via gradient descent, and $\sigma(\cdot)$ denotes the sigmoid activation function to constrain their values within the interval $(0, 1)$.

To ensure the stability and adaptability of the prior weight computation, the coefficients α and β are designed as trainable scalars. In implementation, they are realized through `nn.Parameter` in PyTorch and optimized jointly with all other network parameters by standard backpropagation, rather than being updated in a staged manner. This joint optimization enables the model to automatically adjust the balance between the global semantic guidance and the pixel-level consistency during training. Furthermore, since the majority of pixels in CD tasks usually belong to unchanged regions, the sigmoid normalization of α and β within $(0, 1)$ effectively prevents numerical overflow when $D^{i(changed)}$ becomes dominant. Consequently, the semantic prior weights remain responsive to minority changed regions without causing gradient instability or bias toward the unchanged class. Empirically, the values of α and β converge to moderate magnitudes after training, demonstrating that the learning process is stable under class-imbalance conditions.

Finally, the 1-D weight matrix W_j^i is elementwise multiplied with the original feature blocks along corresponding dimensions, effectively enhancing features in *changed* regions while suppressing those in *unchanged* areas. This mechanism ensures that the LSM adaptively focuses on discriminative features during attention computation.

3) Gaussian Filter-Based Frequency Separation: In digital image processing, a common approach is to transform images from the spatial domain to the frequency domain for manipulation, then reconstruct them back to the spatial domain via inverse transforms. Common frequency-domain filters include ideal filters, Butterworth filters, and Gaussian filters; among these, Gaussian filtering [38] is particularly popular due to its smoothness and differentiability. Compared with purely spatial convolution, the fast Fourier transform (FFT) [39] efficiently converts a 2-D $H \times W$ pixel representation into a complex frequency representation without losing global structural information.

Therefore, we employ FFT combined with Gaussian filtering to leverage FFT's precise spectral decomposition and the Gaussian function's smooth frequency-response characteristics, enabling compact frequency-domain feature separation through elementwise multiplication rather than heavy spatial convolution, thus forming a theoretically grounded yet lightweight framework.

Gaussian filtering in the frequency domain can be implemented in two symmetric forms: LP and HP. The LP mask is defined as

$$\mathrm{H}_{\mathrm{LP}}(u, v) = \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right) \tag{7}$$

where (u, v) are frequency coordinates, and σ is the standard deviation controlling the scale of retained low-frequency content. The corresponding HP mask is

$$\mathrm{H}_{\mathrm{HP}}(u, v) = 1 - \exp\left(-\frac{u^2 + v^2}{2\sigma^2}\right). \tag{8}$$

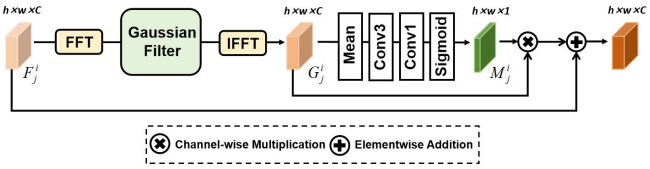


Fig. 6. Illustration of the GFS.

Clearly, these two masks are complementary: when $(u^2 + v^2)^{1/2}$ is small, the LP mask approaches 1, and the HP approaches 0; at large $(u^2 + v^2)^{1/2}$, the opposite holds. Thus, a single parameter σ flexibly enables either global low-frequency enhancement or local high-frequency detail extraction.

Therefore, we design the GFS module, as shown in Fig. 6. First, Gaussian filtering is applied to the input feature map F_j^i to extract either low-frequency or high-frequency information. Then, a channel compression operation followed by a sigmoid activation is used to generate a weight mask M_j^i, which adaptively adjusts the contribution of the filtered feature G_j^i via elementwise multiplication. Finally, the result is fused with the original input F through a residual connection, producing features enhanced with global contours or local details. Through this design, GFS not only preserves the basic information of the original pixels but also dynamically enhances global scene dependencies or local distributions based on task demands.

In the MCA module, the GFS that employs Gaussian LP filtering is formally termed LP-GFS, while the GFS utilizing Gaussian HP filtering is officially named HP-GFS.

From a theoretical perspective, HP and LP Gaussian filters correspond to complementary subspaces in the spectral domain. The HP component emphasizes edges, texture variations, and fine-scale local changes; such boundary cues are especially salient in high-resolution RS imagery, where local dependencies between adjacent objects (e.g., roads and buildings) are frequently reflected by boundary co-location and geometric continuity. In contrast, the LP component retains large-scale consistency and suppresses high-frequency noise. Processing these two components separately enables the network to model local–global complementarity and reduces mutual interference between high- and low-frequency information during attention computation.

This frequency factorization is more principled than direct spatial or frequency-domain feature fusion, because it decomposes the input into orthogonal frequency bands that correspond to distinct physical meanings of change—fine local details versus coarse global context. Empirically, as shown by the w/o GFS variant in Section V, removing this factorization and instead performing direct feature fusion leads to a clear drop in performance, indirectly validating the theoretical advantage of the proposed HP–LP separation design.

D. Joint Loss Function

Based on our comparative experiments, the model achieves the best performance when trained with the cross-entropy (CE) loss, which is defined as follows:

$$\mathcal{L}_{CE} = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right]. \tag{9}$$

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TABLE I
QUANTITATIVE COMPARISONS ON THE THREE CD TEST SETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

Method	LEVIR-CD IoU / F1 / Pre / Rec	WHU-CD IoU / F1 / Pre / Rec	GZ-CD IoU / F1 / Pre / Rec
FC-EF [13]	83.43 / 89.62 / 90.04 / 89.20	79.52 / 87.38 / 88.34 / 86.48	76.24 / 85.21 / 88.13 / 82.81
IFNet [16]	87.75 / 91.95 / 93.14 / 90.83	87.05 / 91.91 / 92.74 / 91.10	81.73 / 87.92 / 90.42 / 89.44
SNUNet [20]	87.42 / 92.35 / 93.17 / 91.56	88.05 / 93.28 / 91.33 / 93.47	82.30 / 88.97 / 90.80 / 89.17
ChangeFormer [25]	87.65 / 92.29 / 93.18 / 91.44	86.02 / 91.97 / 93.03 / 90.97	80.33 / 87.64 / 91.72 / 86.07
DMINet [23]	87.97 / 92.08 / 93.92 / 91.28	88.31 / 93.45 / 91.73 / **94.33**	82.09 / 88.78 / 92.99 / **92.57**
SEIFNet [33]	87.41 / 91.75 / 93.40 / 91.13	85.49 / 91.63 / 89.33 / 94.24	81.30 / 88.27 / 93.12 / 90.27
ISDANet [42]	87.53 / 92.22 / 93.45 / 91.99	80.81 / 90.35 / 83.38 / 90.48	80.57 / 88.48 / 84.89 / 91.51
GCFENet	**88.65** / **93.68** / **94.69** / **92.72**	**89.52** / **94.19** / **94.21** / 94.18	**82.45** / **89.72** / **93.56** / 86.63

During loss computation, we first merge the feature blocks embedded with semantic prior and introduce self-learned scalar parameters to balance the contributions of four-layer feature maps, with a single softmax normalization across stages to ensure nonnegative weights satisfying $\sum_i^4 w_i^{prior} = 1$. The same adaptive weighting policy is also used in the final prediction head, as described in Section III-A, to maintain consistent fusion behavior throughout the network. This produces a preclassification representation Prior for changed and unchanged classes. The CE loss is then applied to both the preclassification representation ($\mathcal{L}_{prior}$) and the final pixelwise prediction ($\mathcal{L}_{pixel}$), with a coefficient of λ assigned to $\mathcal{L}_{prior}$. The value of λ is 0.6. The final loss function of the model is defined as

$$\mathcal{L} = \lambda \cdot \mathcal{L}_{prior} + \mathcal{L}_{pixel}. \tag{10}$$

This design ensures that the model optimizes both the semantic consistency of preclassification features and the fine-grained accuracy of pixelwise predictions, enhancing overall performance in complex change scenarios.

IV. EXPERIMENTS

In this section, we introduce the experimental details and result analysis. Section IV-A presents the three CD datasets used in this experiment. Section IV-B describes the training configurations and evaluation metrics. Section IV-C compares the proposed model with state-of-the-art (SOTA) methods and discusses the experimental results. Section IV-D compares GCFENet with previous models in terms of the model parameters (Params.), floating-point operations per second (FLOPs), and the training time (Time).

A. Datasets

Three representative CD datasets are selected to validate the practical performance of GCFENet.

1) *LEVIR-CD [2]:* As a large-scale RS building CD dataset, it contains 637 high-resolution Google Maps image patch pairs with a spatial size of 1024 × 1024. The changed buildings include diverse types, such as villa residences, apartments, and large warehouses. Images are cropped into 256 × 256 nonoverlapping patches, yielding 7120, 1024, and 2048 image pairs for training, validation, and testing sets, respectively.
2) *WHU-CD [40]:* This dataset, recorded by Wuhan University, captures post-2011 earthquake building constructions in Christchurch, New Zealand. It consists of a pair of aerial images with a spatial scale of 32 507 × 15 354. Original images are cropped into 256 × 256 patches and divided into training/validation/test sets in a 8/1/1 ratio.
3) *GZ-CD [41]:* As a VHR satellite image CD dataset, it focuses on the suburban areas of Guangzhou City, China, spanning from 2006 to 2019. It contains 19 pairs of seasonally varying images. The original images are uniformly cropped to a size of 256 × 256. These image patches are further divided into 2504 training pairs, 313 validation pairs, and 313 test pairs, providing a structured dataset for CD research in suburban environments.

B. Training Details and Evaluation Metrics

1) *Implementation Details:* The GCFENet model is implemented using the PyTorch framework and trained on an NVIDIA RTX 3090 GPU. We adopt the AdamW optimizer with a batch size of 8, an initial learning rate of $1e^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay of 0.01. All models are trained for 200 epochs with identical data augmentation strategies during the training phase. Unless otherwise noted, the Gaussian bandwidth in GFS is fixed to $\sigma = 2$ for all stages.
2) *Evaluation Metrics:* Four widely used metrics are employed to quantitatively assess the performance of CD: intersection over union (IoU), $F1$-score ($F1$), Precision (Pre), and Recall (Rec). The definitions are as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{11}$$

$$F1 = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{12}$$

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{13}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{14}$$

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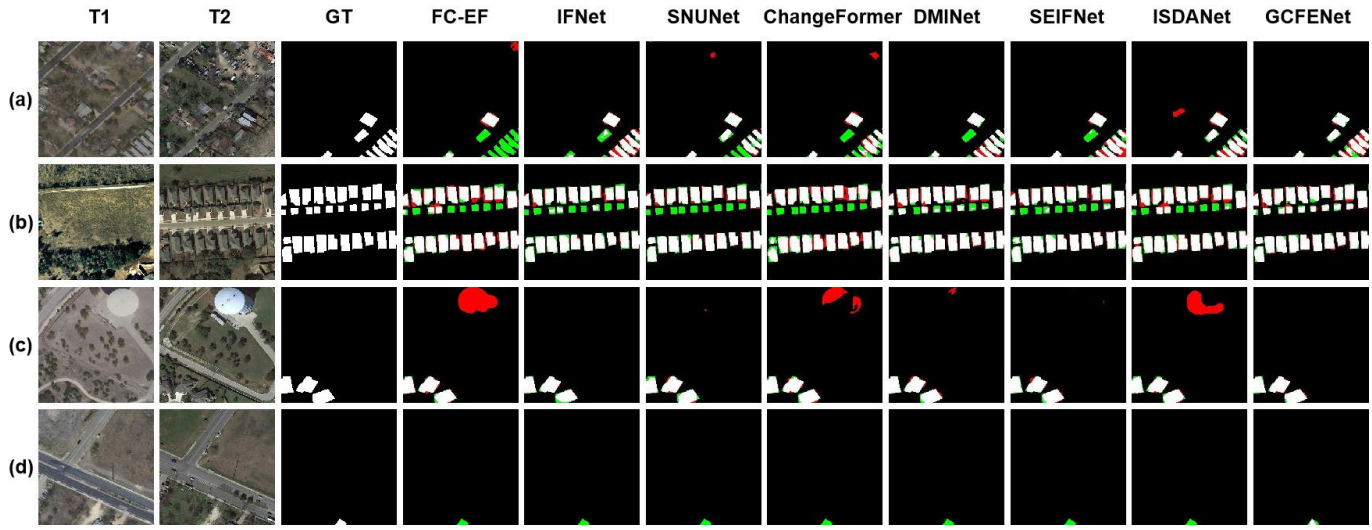


Fig. 7. Visualization results of different methods on the LEVIR-CD test set. (a)–(d) Representative samples. White indicates a correct positive prediction, black signifies a correct negative prediction, red represents an incorrect positive prediction, and green denotes an incorrect negative prediction.

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

C. Comparison With SOTA Methods

We compared the proposed GCFENet with several representative methods, namely FC-EF [13], IFNet [16], SNUNet [20], ChangeFormer [25], DMINet [23], SEIFNet [33], and ISDANet [42].

1) FC-EF [13] is a model based on a pure convolutional U-Net, which concatenates a pair of input images in the channel dimension and then feeds them into the network.
2) IFNet [16] fuses the features of bi-temporal images based on an attention mechanism and improves detection accuracy with the aid of a deep supervision mechanism.
3) SNUNet [20] is a densely connected siamese network model, which strengthens feature propagation through a dense connection mechanism to improve detection accuracy.
4) ChangeFormer [25] extracts features of temporal RS images through hierarchical transformer encoders and uses an MLP decoder to identify changed regions.
5) DMINet [23] realizes CD by extracting multilayer features of bi-temporal RS images and aggregating differences through a dual-branch structure and a cross-temporal joint attention mechanism.
6) SEIFNet [33] is a lightweight RS image CD model using a Siamese backbone, employing ST-DEM and ACFM to capture spatiotemporal differences and multiscale features, suppressing pseudo-changes and scale variations to enhance feature representation for accurate detection.
7) ISDANet [42] fuses multiscale semantic features, integrates self- and cross-attention for deep bi-temporal feature interaction, and employs a supervisory attention module to aggregate hierarchical features during decoding.

For a fair comparison, all methods are trained under the same conditions based on the officially released source codes.

TABLE II
COMPARISON RESULTS ON MODEL EFFICIENCY

Model	Params. (M)	FLOPs (G)	Time (Min)
FC-EF [13]	1.35	3.57	0.83
IFNet [16]	50.71	41.1	7.11
SNUNet [20]	1.35	4.72	6.34
ChangeFormer [25]	41.02	202.78	3.43
DMINet [23]	6.24	14.55	4.01
SEIFNet [33]	27.91	8.37	1.71
ISDANet [42]	6.93	3.46	4.20
GCFENet	115.3	8.72	4.94

1) Quantitative Evaluation: Numerically, Table I shows the overall performance of all methods on the LEVIR-CD, WHU-CD, and GZ-CD sets. The evaluation metrics include IoU, $F1$, Pre, and Rec. Bold numbers represent the best results. Evidently, the GCFENet exhibits superior performance in comparison to alternative methods. On three datasets, GCFENet achieved the highest IoU and $F1$ results. Specifically, on the LEVIR-CD dataset, GCFENet's IoU and $F1$ reached 88.65% and 93.68%, respectively, exceeding the best-performing competing methods by 0.68% in IoU and 1.33% in $F1$. On the WHU-CD dataset, GCFENet obtained IoU and $F1$ values of 89.52% and 94.19%, outperforming competitors by 1.21% in IoU and 0.74% in $F1$. For the GZ-CD dataset, GCFENet achieved 82.45% in IoU and 89.72% in $F1$, surpassing state-of-the-art methods by 0.15% and 0.75% in these metrics, respectively. This sufficiently demonstrates the effectiveness of our model, which is tailored to the unique object relationships in RS images. In addition, the slightly lower Precision or Recall of GCFENet on certain datasets may be attributed to the imbalanced distribution of positive and negative samples in the test sets of these datasets.

2) Qualitative Evaluation: To further illustrate the validity of our proposed method, qualitative analyses are conducted on LEVIR-CD, WHU-CD, and GZ-CD test sets (see Figs. 7–9). Distinct colors are used to represent the accuracy of

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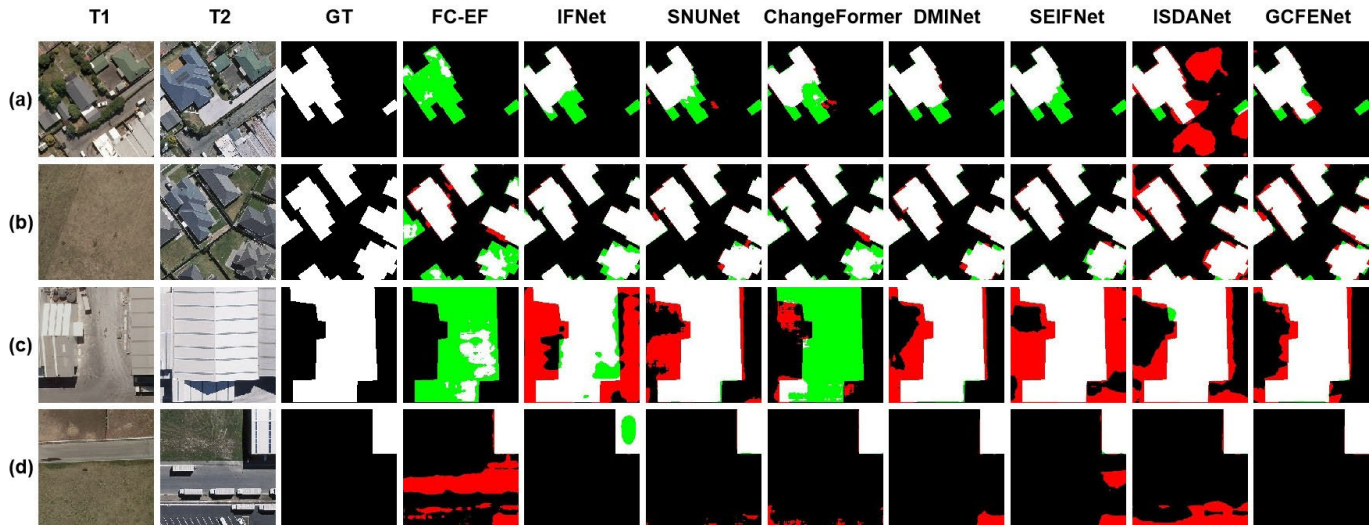


Fig. 8. Visualization results of different methods on the WHU-CD test set. (a)–(d) Representative samples. White indicates a correct positive prediction, black signifies a correct negative prediction, red represents an incorrect positive prediction, and green denotes an incorrect negative prediction.

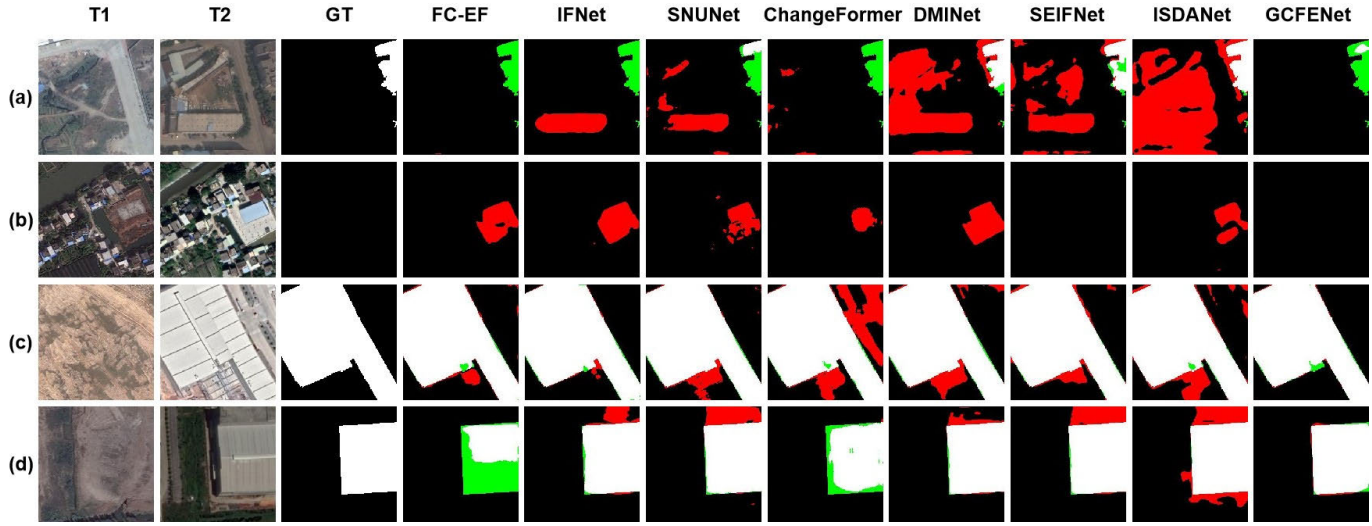


Fig. 9. Visualization results of different methods on the GZ-CD test set. (a)–(d) Representative samples. White indicates a correct positive prediction, black signifies a correct negative prediction, red represents an incorrect positive prediction, and green denotes an incorrect negative prediction.

detection outcomes: true positives (TP) are marked in white, true negatives (TN) in black, false positives (FP) in red, and false negatives (FN) in green.

a) Visualization on LEVIR-CD (see Fig. 7): Four representative change samples were selected for visual analysis. Fig. 7(a) captures partial disappearance and new emergence of building clusters, while Fig. 7(b) shows neatly arranged houses constructed on vacant land. Fig. 7(c) and (d) depicts small huts built alongside roads. As shown in Fig. 7, GCFENet more accurately identifies newly added structures in Fig. 7(a) without false detections of vanished buildings. In Fig. 7(b), GCFENet, similar to prior models, effectively recognizes large-scale new constructions and demonstrates superior performance in detecting small building clusters. Fig. 7(c) and (d) further illustrates that GCFENet, designed based on object relationships in RS images, leverages scene features (e.g., roads) to achieve excellent CD results. In summary, the qualitative analysis aligns with the quantitative results in Table I, confirming that the proposed GCFENet achieves the best performance on the LEVIR-CD dataset.

b) Visualization on WHU-CD (see Fig. 8): Like the LEVIR-CD dataset, Fig. 8(a) and (c) in WHU-CD illustrate building turnover within clustered areas, Fig. 8(b) depicts large-scale construction on extensive vacant land, and Fig. 8(d) shows new buildings emerging alongside roads. Compared with prior methods, our GCFENet achieves more accurate CD with fewer errors. Notably, its prediction map for Fig. 8(d) aligns nearly perfectly with the ground truth (GT). Thus, both quantitative and qualitative analyses confirm that GCFENet performs exceptionally well on the WHU-CD dataset.

c) Visualization on GZ-CD (see Fig. 9): Among the four representative samples in the GZ-CD dataset, Fig. 9(a) depicts new buildings emerging alongside roads, Fig. 9(b) shows the same area at different times with no actual structural changes (i.e., pseudo-change), and Fig. 9(c) and (d) captures large-scale construction on extensive barren land. As illustrated in

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TABLE III
ABLATION STUDIES ON DIFFERENT MODULES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. ALL RESULTS ARE DESCRIBED AS PERCENTAGES (%)

Module	LEVIR-CD IoU / F1 / Pre / Rec	WHU-CD IoU / F1 / Pre / Rec	GZ-CD IoU / F1 / Pre / Rec
w/o DFI	85.70 / 91.80 / 91.49 / 92.12	75.36 / 84.09 / 82.05 / 86.44	79.27 / 87.48 / 90.62 / 84.90
w/o LSM	86.89 / 92.56 / 94.79 / 90.57	87.16 / 92.71 / 92.19 / 93.25	79.60 / 87.99 / 90.97 / 85.52
w/o GFS	87.27 / 92.44 / 93.82 / 91.16	85.11 / 91.37 / 91.69 / 91.05	80.56 / 88.40 / 91.90 / 85.56
w/o MCA	86.10 / 91.70 / 93.91 / 92.21	83.89 / 90.54 / 87.92 / 93.60	79.13 / 87.38 / 90.08 / 85.10
GCFENet	**88.65 / 93.68 / 94.69 / 92.72**	**89.52 / 94.19 / 94.21 / 94.18**	**82.45 / 89.72 / 93.56 / 86.63**

Fig. 9, most models yield suboptimal results for Fig. 9(a), yet GCFENet exhibits fewer errors. For Fig. 9(b), the majority of models misclassify the pseudo-change, whereas only SEIFNet [33] and GCFENet remain error-free. Fig. 9(c) and (d) further demonstrates the remarkable precision of GCFENet's predictions. In summary, the visual analysis and qualitative results collectively underscore the exceptional performance of GCFENet.

D. Model Efficiency

To further evaluate the efficiency of our proposed model, we compare it with several state-of-the-art methods in terms of model parameters (Params.), FLOPs, and the training time (Time) for one epoch on LEVIR-CD, with an input size of $256 \times 256 \times 3$. As shown in Table II, our model has a significantly higher number of parameters (115.3M) than most existing methods, reflecting its greater representational capacity. However, despite the large parameter size, our model achieves a relatively low computational cost with only 8.72 GFLOPs, which is notably lower than ChangeFormer [25] (202.78 GFLOPs) and IFNet [16] (41.1 GFLOPs), indicating the computational efficiency of our architecture.

Although the parameter count of GCFENet is considerably large, most of the additional weights come from lightweight 1×1 projection layers and frequency-domain gating branches introduced to model multiscale contextual relations. These components mainly perform channelwise transformations and contain negligible spatial kernels, so they have little impact on computational cost. Consequently, the overall FLOPs and memory usage remain close to those of previous methods, while the added parameters increase the representational flexibility of the network and lead to consistent performance gains.

Furthermore, our training time per epoch is 4.94 min, which is competitive compared to models with similar or fewer parameters, such as SNUNet [20] (6.34 min) and IFNet [16] (7.11 min). This demonstrates that our model strikes a good balance between complexity and efficiency, benefiting from an optimized network design that facilitates fast training while maintaining high modeling capacity. The results confirm that our method is computationally efficient and scalable for practical applications in CD tasks.

V. ABLATION STUDIES

To validate the effectiveness of individual modules and the coefficient in the loss function, ablation experiments were conducted on three datasets.

TABLE IV
ABLATION STUDY OF DFI VARIANTS ON THE LEVIR-CD DATASET

Variant	IoU (%)	F1 (%)
w/o DFI	85.70	91.80
w/o Add	86.41	92.53
GCFENet	**88.65**	**93.68**

A. Ablation Studies on Modules

In this section, as shown in Table III, we conducted ablation experiments on these innovative modules. Figs. 10–13 show the visual results of ablation experiments on the LEVIR-CD dataset.

1) Effectiveness of DFI: To assess the effectiveness of the proposed DFI module, we conducted ablation experiments by replacing it with a simple pixelwise subtraction operation. As shown in Table III, removing DFI leads to a significant performance degradation across all datasets. On LEVIR-CD, the IoU and $F1$-score dropped by 2.95% and 1.88%, respectively. On WHU-CD, the decrease was even more pronounced, with IoU and $F1$ dropping by 14.16% and 10.1%. On GZ-CD, we observed a 3.18% reduction in IoU and a 2.24% drop in $F1$. These results demonstrate that DFI is more effective than direct pixel subtraction in capturing meaningful semantic differences between bi-temporal images, especially in complex scenarios with noisy backgrounds or subtle structural changes.

Considering that subtraction is the core operation for change modeling, we also conducted an additional ablation by removing only the addition branch while retaining the subtraction pathway. This variant, denoted as *w/o Add*, verifies the complementary role of the addition branch in enhancing global semantic consistency. As shown in Table IV, removing the addition branch results in a mild but consistent performance drop, confirming that the additive pathway contributes to stabilizing large-scale context and complements the subtraction branch that focuses on local change details.

In addition to the quantitative evaluation, we also provide qualitative comparisons in Fig. 10. The visualizations are selected from the LEVIR-CD dataset and include three representative examples. In Fig. 10(a), without the assistance of DFI, the model incorrectly identifies unchanged regions as changed, resulting in a high number of false positives (highlighted in red). As shown in Fig. 10(b) and (c), our DFI module significantly enhances the model's ability to

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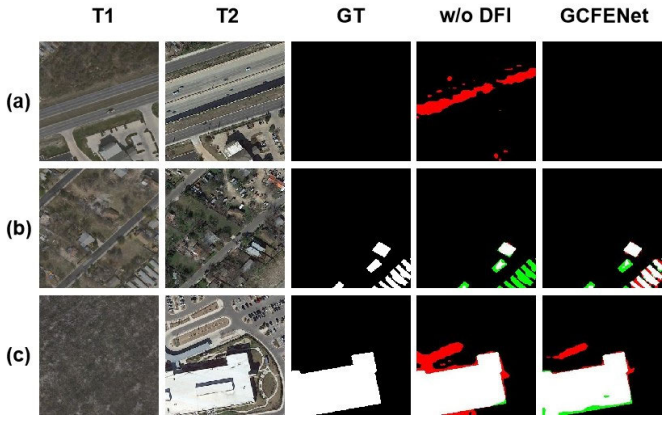


Fig. 10. Visual ablation of module DFI on LEVIR-CD: w/ versus w/o.

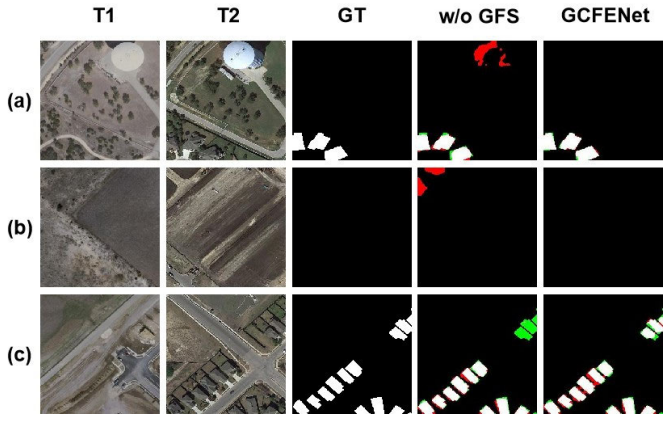


Fig. 12. Visual ablation of module GFS on LEVIR-CD: w/ versus w/o.

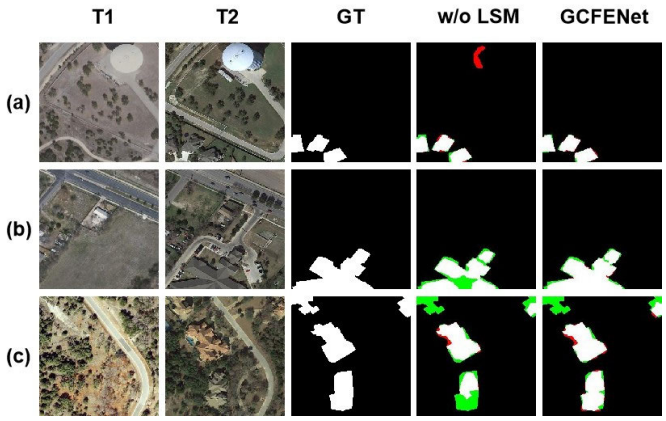


Fig. 11. Visual ablation of module LSM on LEVIR-CD: w/ versus w/o.

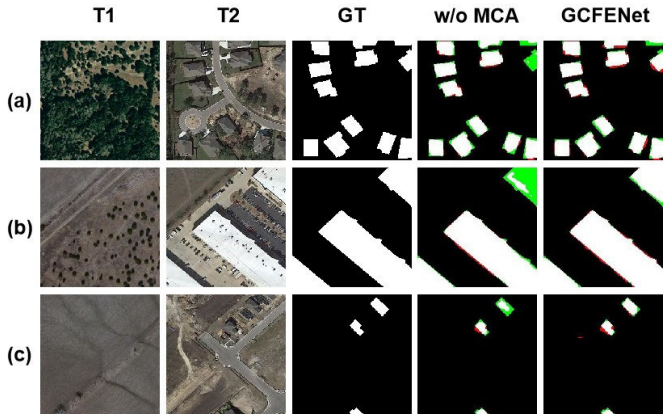


Fig. 13. Visual ablation of module MCA on LEVIR-CD: w/ versus w/o.

capture fine-grained change boundaries and subtle edge details. These improvements further confirm the critical role of DFI in facilitating accurate and robust CD.

2) Effectiveness of LSM: Similarly, we ablated the LSM module by directly feeding the patchwise features into the key branch of the attention mechanism, thereby removing the semantic prior fusion strategy. As shown in Table III, this modification leads to a consistent performance degradation across all datasets. On LEVIR-CD, IoU and $F1$-score dropped by 1.76% and 1.12%, respectively. On WHU-CD, the drops were 2.36% and 1.48%, and on GZ-CD, 2.85% and 1.73%, respectively. While the decline is not as dramatic as when removing DFI, these results clearly indicate that the semantic prior fusion strategy provides valuable global context, helping the model better distinguish real changes from background noise.

To further illustrate this, we provide qualitative comparisons on LEVIR-CD in Fig. 11. In Fig. 11(a), the model without semantic prior fusion mistakenly detects unchanged regions as changes (false positives, red). Fig. 11(b) and (c) reveals that the lack of high-level semantic guidance impairs the model's ability to precisely localize change boundaries, resulting in fragmented and noisy predictions (false negatives in green). These visual results confirm that incorporating contextual priors enhances the model's spatial awareness and robustness, particularly in scenes with ambiguous textures or low-contrast changes.

3) Effectiveness of GFS: To evaluate the contribution of the GFS module, we conducted ablation experiments by removing it from both the query and value branches. In the ablated version, the attention mechanism directly receives the patchwise feature maps without any structural enhancement. As shown in Table III, removing these modules leads to a noticeable performance drop across all datasets. On LEVIR-CD, the IoU and $F1$-score dropped by 1.38% and 1.24%, respectively; on WHU-CD, these metrics fell by 4.41% and 2.82%; and on GZ-CD, they decreased by 1.89% and 1.32%.

Qualitative results are shown in Fig. 12. In Fig. 12(a), the model yields substantial false positives, detecting changes that do not exist. In Fig. 12(b), false alarms appear in unchanged regions, indicating poor background suppression. In Fig. 12(c), many true changes are missed, reflecting a significant increase in false negatives. These results confirm that the GFS module enhances the spatial discrimination of attention and contributes to more reliable change localization.

We further include two representative query-side explicit spatial encodings—learnable absolute positional embedding (LAPE) [27] and convolutional positional encoding (CPE) [43]—as like-for-like replacements of HP-GFS in the query branch. As reported in Table V, neither of these explicit encodings improves performance; instead, the metrics drop slightly

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TABLE V

COMPARISON WITH EXPLICIT POSITIONAL ENCODINGS ON THE LEVIR-CD DATASET BY REPLACING HP-GFS IN THE QUERY BRANCH. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Query Encoding	IoU / F1 / Pre / Rec
LAPE [27]	88.28 / 93.38 / 94.51 / 92.31
CPE [43]	88.21 / 93.36 / **95.03** / 91.83
HP-GFS(ours)	**88.65** / **93.68** / 94.69 / **92.72**

TABLE VI

ABLATION STUDY ON THE LOSS FUNCTION COEFFICIENTS ON THE LEVIR-CD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Coefficient of $\mathcal{L}_{\text{prior}}$	IoU / F1 / Pre / Rec
0.2	88.08 / 93.32 / 94.31 / 91.52
0.4	88.41 / 93.53 / 94.48 / 92.62
0.6	**88.65** / **93.68** / 94.69 / **92.72**
0.8	88.63 / 93.66 / 95.05 / 92.37
0.0 (only $\mathcal{L}_{\text{pixel}}$)	87.21 / 92.77 / **95.22** / 90.61

and consistently. This suggests that the proposed frequency-based boundary prior is more compatible with our CD setting than standard query-side spatial encoding strategies.

4) Effectiveness of MCA: In addition, we removed the entire MCA module and reverted to a baseline attention design, where patchwise features are directly used as inputs to the query, key, and value branches without any structural enhancement. As shown in Table III, this modification leads to a noticeable performance drop across all datasets. On LEVIR-CD, IoU and $F1$-score decreased by 2.55% and 1.98%, respectively; on WHU-CD, by 5.63% and 3.65%; and on GZ-CD, by 3.32% and 2.34%. These results suggest that the MCA module plays a vital role in enhancing the model's ability to capture diverse spatial dependencies and subtle changes.

Fig. 13 shows qualitative comparisons on the LEVIR-CD dataset. In Fig. 13(a), the prediction becomes cluttered with false positives, especially around object edges. In Fig. 13(b), the model struggles with under-segmentation, resulting in inaccurate change boundaries. In Fig. 13(c), small changes are completely missed, indicating a decline in fine-grained detection capability. Together, these examples demonstrate that the MCA module contributes to clearer, more reliable change localization.

B. Ablation Study on Loss Function

To determine the optimal balance between different supervision signals, we conducted ablation experiments by varying the weight of the preclassification representation loss $\mathcal{L}_{\text{prior}}$ while keeping the coefficient of the final pixelwise prediction loss $\mathcal{L}_{\text{pixel}}$ fixed at 1.0. This set of experiments was conducted on the LEVIR-CD dataset alone. As shown in Table VI, we evaluated four different settings for the coefficient of $\mathcal{L}_{\text{prior}}$: 0.2, 0.4, 0.6, and 0.8. The experimental results show that the best segmentation performance is achieved when the weight of $\mathcal{L}_{\text{prior}}$ is set to 0.6.

TABLE VII

$F1$ (%) ON THE LEVIR-CD DATASET UNDER DIFFERENT σ CONFIGURATIONS

σ	0.5	1	2	3	4	Scale-dep.
F1	93.45	93.47	**93.68**	93.66	93.62	93.57

This indicates that introducing a moderate level of prior supervision helps guide the model's feature learning in the early stages while avoiding overemphasis on preclassification that may hinder the optimization of the final pixelwise predictions. These findings confirm that an appropriate balance between high-level representation learning and fine-grained prediction is crucial for optimal performance.

For completeness, we further include the case where $\mathcal{L}_{\text{prior}}$ is removed (i.e., $\lambda = 0$), showing that the network is supervised solely by the pixel-level loss. As shown in Table VI, this setting leads to a noticeable drop in performance, indicating that the semantic prior loss provides complementary guidance that enhances global semantic consistency and suppresses false detections in ambiguous regions.

C. Sensitivity to σ

To evaluate the robustness of the Gaussian bandwidth, we conducted two sets of experiments on the LEVIR–CD dataset: 1) fixed σ values of $\{0.5, 1, 2, 3, 4\}$ and 2) scale-dependent values proportional to the feature-map size. In the latter configuration, the σ values for LP filtering were $\{12.8, 6.4, 3.2, 1.6\}$ for spatial sizes $\{64 \times 64, 32 \times 32, 16 \times 16, 8 \times 8\}$, while the HP branch used $\{3.2, 1.6, 0.8, 0.4\}$ for $\{16 \times 16, 8 \times 8, 4 \times 4, 2 \times 2\}$.

As shown in Table VII, the variation in performance is within 0.5%, indicating that the proposed GFS is highly insensitive to the specific choice of σ. This robustness stems from the convolutional backbone, which harmonizes multi-scale spatial statistics, and from the relative nature of the Gaussian bandwidth in the frequency domain. Hence, the fixed $\sigma = 2$ setting is adopted for all main experiments.

VI. CONCLUSION

We propose a novel GCFENet, designed to address CD in RS images by capturing global scene dependencies and local distributions, in consideration of the strong geospatial correlations inherent in such data. First, a Siamese pretrained ResNet18 backbone extracts multilevel features, which are then paired and fed into the DFI module to enhance sensitivity to subtle change details while incorporating overall scene information. Next, the MCA module, which comprises a three-branch architecture, reinforces both pixelwise adjacency correlations and broader contextual cues. In addition, the LSM module endows intermediate features with a semantic prior, effectively linking local distributions to global scene dependencies. Finally, the feature map optimized by the MCA module is input to an attention mechanism, fully leveraging attention's capability in the RS domain to yield more precise CD results. Extensive experiments on three public datasets demonstrate that GCFENet outperforms previous state-of-the-art methods, and ablation studies validate the effectiveness of each innovative component. In future work, we aim to further exploit interrelations among multilevel features and develop a more advanced fusion mechanism to enhance performance.

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